

JEE MAIN REVISION MASTER: WAVE OPTICS 2026

1 QUESTIONS 1.67% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 26 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Wave Optics - Core Application	26 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Wave Optics - Core Application (Very High Yield — 26 PYQs)

Core Principle: Key problem pattern in Kcet Wave Optics.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In Young's double slit experiment, how many maxima can be seen on a screen (including central maxima) if $d = (5\lambda / 2)$ (where λ is wavelength)..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "If A is incident plane wave front then refracted wave front in $(n_2 > n_1)$..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Three polaroid sheets are co-axially placed as indicated in the diagram. Pass axes of the polaroids 2 and 3 make 30° and 90° with the pass axis of polaroid 1..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In Young's double slit experiment, how many maxima can be seen on a screen (including central maxima) if $d = (5\lambda / 2)$ (where λ is wavelength of light and d is distance between the two slits).

- (A) 5
- (B) 4
- (C) 7
- (D) 1

Correct Answer: None

Solution Analysis: For bright fringes or maxima in Young's double slit experiment, the path difference must be an integral multiple of wavelength: $d \sin \theta = n\lambda$, where $n = 0, \pm 1, \pm 2, \dots$. Since the maximum possible value of $\sin \theta$ is 1, $|n\lambda| \leq d$. Given, $d = (5\lambda / 2)$. So, $|n| \leq (d / \lambda)$ $|n| \leq (5 / 2)$ $|n| \leq 2.5$. But n must be an integer. So possible values of n are $n = -2, -1, 0, 1, 2$. That gives total maxima: 5. So the correct option is: 5 Option A.